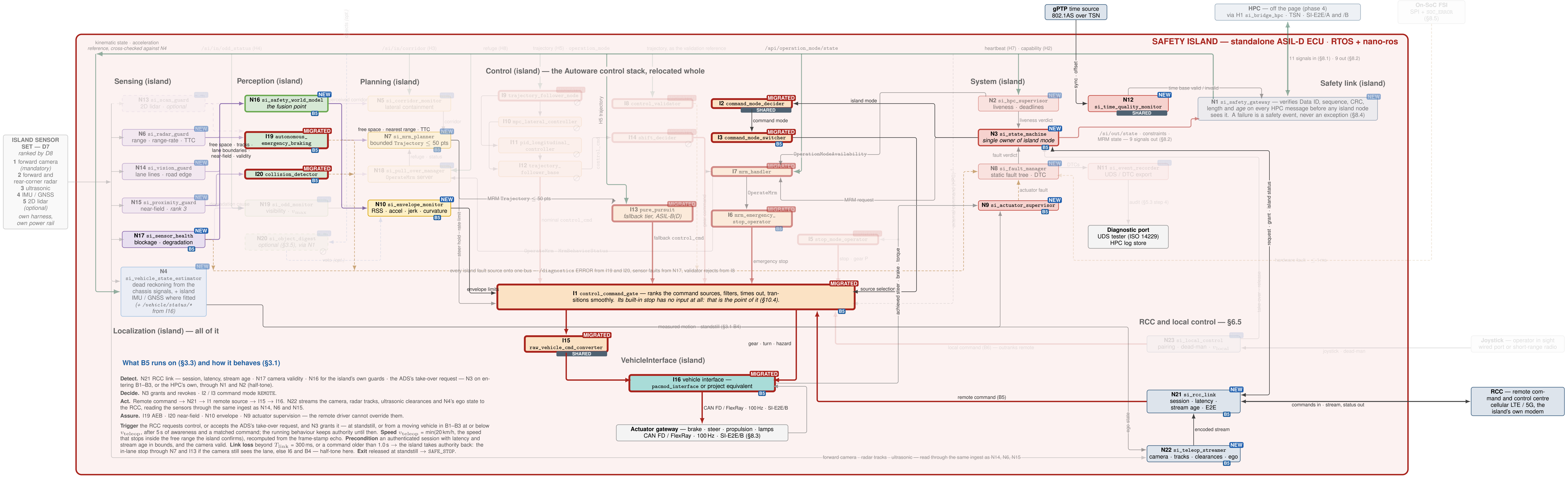
Phase 5 · B5 — Tele-operation from the remote command centre

A remote driver at the RCC drives the vehicle over the island's own communication module, on the camera, radar and ultrasonic the island streams — while the island keeps its guards armed and stops the vehicle itself if the link fails.
The phase-5 drawing, unchanged: what B5 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Detailed design: §6.6 — from standstill, or taken over from a moving vehicle in B1–B3 on the ADS's request.*



gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
objects (opt.)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
sync · offset
11 signals in (§8.1) · 9 out (§8.2)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
N6 si_radar_guard
range · range-rate · TTC
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
NEW
B5
Perception (island)
N16 si_safety_world_model
the fusion point
I19 autonomous_emergency_braking
I20 collision_detector
MIGRATED
N19 si_odd_monitor
visibility · v_{max}
N20 si_object_digest
optional (§3.5), via N1
free space · tracks · lane boundaries · near-field · validity
degradation cause
veto (opt.)
perceived corridor
Planning (island)
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
N10 si_envelope_monitor
RSS · accel · jerk · curvature
v_{max}
steer hold · rate limit
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
I8 control_validator
I14 shift_decider
H5 trajectory
control_cmd
MRM Trajectory ≤ 50 pts
nominal control_cmd
gear command
I13 pure_pursuit
fallback tier, ASIL-B(D)
fallback control_cmd
OperateMrm · MrmBehaviorStatus
I2 command_mode_decider
SHARED
command mode
I3 command_mode_switcher
island mode
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
emergency stop
I5 stop_mode_operator
stop · gear P
external command (HPC) †
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
envelope limits
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
source selection
achieved steer · brake · torque
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
DTCs
actuator fault
N9 si_actuator_supervisor
N12
si_time_quality_monitor
time base valid / invalid
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
request · grant · island status
take-over · release
hardware fault · ≤ 1 ms
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
measured motion · standstill (§3.1 B4)
I15
raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
RCC and local control — §6.5
N23 si_local_control
pairing · dead-man · v_{local}
local command (B6) — outranks remote
Joystick — operator in sight
wired port or short-range radio
joystick · dead-man
remote command (B5)
N21 si_rcc_link
session · latency · stream age · E2E
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
encoded stream
ego state
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B5 runs on (§3.3) and how it behaves (§3.1)

Detect. N21 RCC link — session, latency, stream age · N17 camera validity · N16 for the island's own guards · the ADS's take-over request — N3 on entering B1–B3, or the HPC's own, through N1 and N2 (half-tone).

Decide. N3 grants and revokes · I2 / I3 command mode `REMOTE`.

Act. Remote command → N21 → I1 remote source → I15 → I16. N22 streams the camera, radar tracks, ultrasonic clearances and N4's ego state to the RCC, reading the sensors through the same ingest as N14, N6 and N15.

Assure. I19 AEB · I20 near-field · N10 envelope · N9 actuator supervision — the remote driver cannot override them.

Trigger the RCC requests control, or accepts the ADS's take-over request, and N3 grants it — at standstill, or from a moving vehicle in B1–B3 at or below v_{teleop}, after 5 s of awareness and a matched command; the running behaviour keeps authority until then. **Speed** v_{teleop} = min(20 km/h, the speed that stops inside the free range the island confirms), recomputed from the frame-stamp echo. **Precondition** an authenticated session with latency and stream age in bounds, and the camera valid. **Link loss** beyond T_{link} = 300 ms, or a command older than 1.0 s → the island takes authority back: the in-lane stop through N7 and I13 if the camera still sees the lane, else I6 and B4 — half-tone here. **Exit** released at standstill → `SAFE_STOP`.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B5

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

∅

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→	HPC-sourced, verified by N1	→	actuator feedback
→	island sensor data	→	state · mode · decision
→	control command into the gate	⇢	optional (§3.5)
⇢	fault · `/diagnostics`		

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B5, tele-operation from the remote command centre

full colour — named for B5 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B5's path by §3.1, but not named in §3.3

faded — not used by B5. An edge is drawn at the weaker of its two ends. Only B5's own tag is drawn on each box.